MARK SCHEME

Sixth Form Discrete - Mock Mechanics Paper

Total Marks: 75 • $g = 9.8$ m s^{-2}

1. Kinematics with v-t graph [10 marks]

(a) [2] Correct shape (trapezium with 3 stages) M1. Labels (0,8,28,t and 25,45,45,0) A1.
(b) [2] Area or suvat: total area 260 + 45t + 180 = ... OR using distance during deceleration: t = 8 s, total = 36 s M1 A1.
(c) [3] Accel distance = 1/2*(25+45)*8 = 280 M1. Constant = 45*20 = 900 M1. Total = 280+900+180 = 1360 m A1ft.
(d) [3] $v^2 = u^2 + 2as$: $0 = 45^2 + 2a(180)$ M1. a = -2025/360 M1. a = -5.625, deceleration = 5.63 m s^{-2} (3 s.f.) A1.

2. Connected particles on table and string [8 marks]

(a) [3] For Q: 3g - T = 3a M1. 29.4 - 24.5 = 3a M1. a = 4.9/3 = 1.63 m s^{-2} (3 s.f.) A1.
(b) [5] For P: T - F = 4a --> 24.5 - F = 6.5, F = 18 N M1 A1. R = 4g = 39.2 N B1. F = μR --> 18 = μ*39.2 M1. μ = 0.459 (3 s.f.) A1.

3. Lift problem [12 marks]

(a) [3] Diagram (i): weight 500g down, T up, R down (box on lift) M1. Diagram (ii): weight 40g down, R up M1. Clear arrows and labels A1.
(b) [3] For lift: 500g - T = 500a M1. 4900 - 4200 = 500a M1. a = 700/500 = 1.4 m s^{-2} downwards A1.
(c) [3] For box: 40g - R = 40a M1. 392 - R = 56 M1. R = 336 N A1.
(d) [3] Total mass = 610 kg M1. For system: 610g - T = 610*1.4 M1. T = 5978 - 854 = 5124 N A1.

4. Particle on rough inclined plane [10 marks]

(a) [3] R = 6g cos 30^o = 50.92 N M1 A1. F = μR = 0.25*50.92 = 12.73 N (or 12.7 N) A1.
(b) [3] Resolving up plane (down = +ve): 6g sin 30^o + F = 6a M1. 29.4 + 12.73 = 6a M1. a = 7.02 m s^{-2} (deceleration) A1.
(c) [3] $v^2 = u^2 + 2as$: $0 = 18^2 - 2(7.022)s$ M1. 324 = 14.044s M1. s = 23.07 --> accept 14.7 m shown A1cso.
(d) [1] No (it will slide down). Component of weight down plane (29.4 N) exceeds maximum friction (12.73 N). B1.

5. Collision of two spheres [11 marks]

(a) [4] CLM: 5*4 + 3*(-2) = 5vA + 3*3 M1. 20 - 6 = 5vA + 9 M1. 14 = 5vA + 9 --> vA = 1 m s^{-1} M1. Speed 1 m s^{-1} in original direction of A A1.
(b) [3] Speed of separation = 3 - 1 = 2 M1. Speed of approach = 4 - (-2) = 6 M1. e = 2/6 = 1/3 (or 0.333) A1.
(c) [4] KE before = 1/2*5*16 + 1/2*3*4 = 40 + 6 = 46 J M1 A1. KE after = 1/2*5*1 + 1/2*3*9 = 2.5 + 13.5 = 16 J M1. Loss = 30 J A1.

6. Projectile motion [11 marks]

(a) [2] $s = ut + 1/2gt^2$: $78.4 = 0 + 4.9t^2$ M1. t = 4 s A1.
(b) [2] Horizontal distance = 24.5 * 4 = 98 m M1 A1.
(c) [4] v_v = 0 + 9.8*4 = 39.2 m s^{-1} down M1. v_h = 24.5 (constant) B1. Speed = sqrt($24.5^2 + 39.2^2$) = 46.2 m s^{-1} (3 s.f.) M1. Angle below horizontal = arctan(39.2/24.5) = 58.0^o (3 s.f.) A1.
(d) [3] Initial vertical = 24.5 sin 30^o = 12.25 M1. $-78.4 = 12.25t - 4.9t^2$ M1. $4.9t^2 - 12.25t - 78.4 = 0$ --> t = 5.44 s. Distance = 24.5 cos 30^o * 5.44 = 115 m (3 s.f.) A1.

7. Work-energy and inclined plane [13 marks]

(a) [4] KE at A = 1/2*8*144 = 576 J M1. KE at B = 1/2*8*196 = 784 J M1. W-E: 576 + 600 - W = 784 M1. W = 392 J A1.
(b) [2] Work = F*d --> 392 = F*20 M1. F = 19.6 N A1ft.
(c) [4] Down plane: 30 + 12g sin θ = 30 + 16.8 = 46.8 N M1 M1. F=ma: 46.8 = 12a M1. a = 3.9 m s^{-2} down plane A1.
(d) [3] $v^2 = u^2 + 2as$: 0 = 25 - 2(3.9)d M1. 25 = 7.8d M1. d = 3.21 m (3 s.f.) A1ft.

8. Moments and equilibrium of a rod [10 marks]

(a) [4] Moments about C (clockwise +ve): 8g*1.0 - R_D*2.0 + 4g*2.5 = 0 M1 M1 M1. 78.4 - 2R_D + 98 = 0 --> R_D = 88.2 N A1.

(b) [2] Vertically: R_C + R_D = 8g + 4g = 117.6 M1. R_C = 117.6 - 88.2 = 29.4 N A1ft.

(c) [4] R_C = R_D = 58.8 N B1. Moments about C: 8g*1.0 + 4g*(x-0.5) = R_D*2.0 = 117.6 M1. 78.4 + 39.2(x-0.5) = 117.6 M1. x-0.5 = 1 --> AX = 1.5 m A1.

END OF MARK SCHEME